

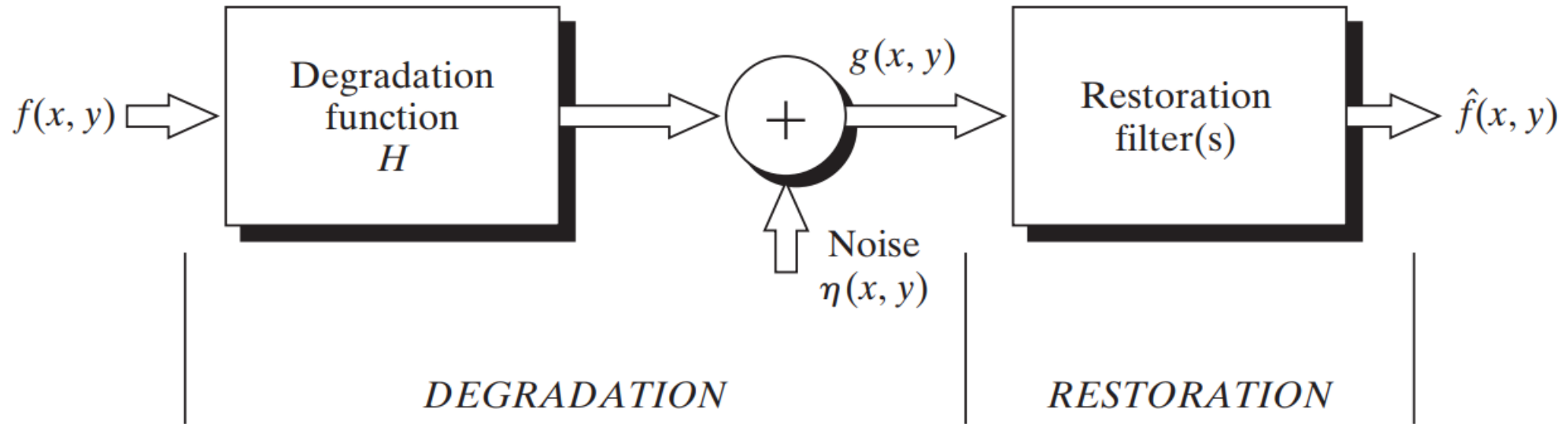
Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-10: Restoration

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-5)

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Degradation Model



Mathematical expression of the degradation operation can be written as the following forms:

1. $g(x, y) = h(x, y) * f(x, y) + \eta(x, y)$
2. $G(u, v) = H(u, v)F(u, v) + N(u, v)$
3. $g = Hf + n$

9.4 Estimation of Degradation Model/ Function:

- ❖ Observation
- ❖ Experimentation
- ❖ Mathematical modeling

9.4.1 Estimation by Image Observation:

(Estimate the degradation function by observation)

Select a Small Region of Interest (ROI)

- ☐ Choose a small section of the degraded image that contains simple structures (e.g., an object and its background).

This region should have strong signal content (high contrast or clear edges) to minimize the impact of noise.

Construct an Unblurred Image of the ROI

Using our knowledge of the scene or assumptions about the original image, create an unblurred version of the selected ROI.

□ For example, if the ROI contains a sharp edge (e.g., an object against a background), we can assume the original edge was a step function.

This unblurred image represents what the ROI should look like without degradation.

Compare the Degraded and Unblurred ROI

$g(x, y)$ = degraded ROI (observed).

$f(x, y)$ = unblurred ROI (constructed).

$$g(x, y) = H[f(x, y)] + \eta(x, y)$$

H = degradation function (e.g., blur kernel).

$\eta(x, y)$ = additive noise.

Estimate the Degradation Function (H)

If noise is negligible $\eta(x,y) \approx 0$, the degradation function H can be estimated by solving:

$$g(x, y) = H[f(x, y)]$$

In the frequency domain, this becomes:

$$G(u, v) = H(u, v) \cdot F(u, v)$$

$$H(u, v) = \frac{G(u, v)}{F(u, v)}$$

$G(u, v)$ = Fourier transform of $g(x, y)$.

$F(u, v)$ = Fourier transform of $f(x, y)$.

$H(u, v)$ = Fourier transform of the degradation function.

9.4.2 Estimation by Experimentation:

Recreate the Imaging Setup

Set up the imaging system (camera, lens, lighting, etc.) in the **same conditions** as when the degraded image was captured.

This ensures that the degradation function H we estimate is accurate for the given setup.

Use a Known Input (Impulse)

Introduce a **known input** into the system. The most common choice is an **impulse** (a point source of light or a bright dot on a dark background).

In image processing, an *impulse is represented as a single bright pixel* (or a small bright spot) in an otherwise dark image.

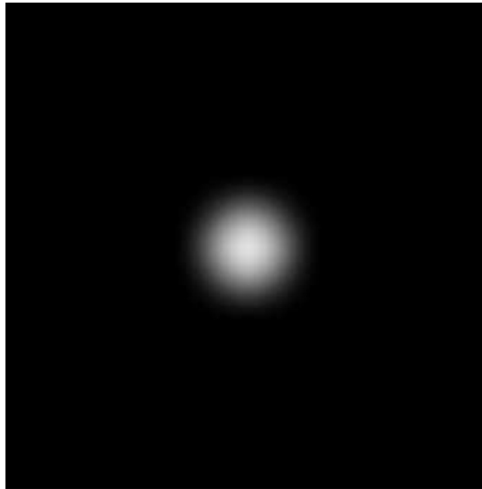
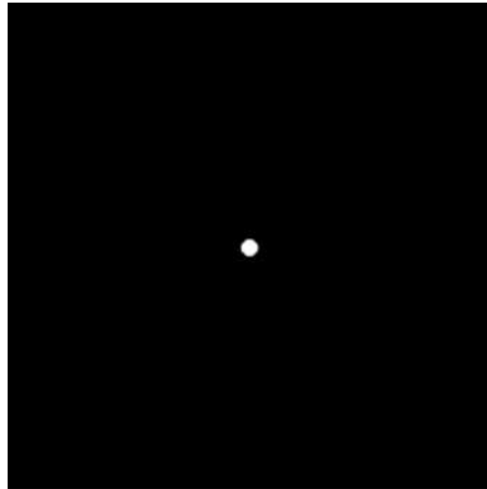
Capture the System's Response

Capture the image of the impulse using the same imaging setup.

The resulting image is called the **Point Spread Function (PSF)** or **impulse response** of the system.

The PSF represents how the imaging system blurs a single point of light, and it is essentially the degradation function H .

(a) An impulse of light (shown magnified).



(b) Imaged (degraded) impulse.

Analyze the PSF

The PSF can be analyzed in the **spatial domain** (as an image) or in the **frequency domain** (using Fourier Transform).

Let, original image, $f(x,y)$ is an impulse and Fourier transform of the impulse is $F(u,v)$.

$$F(u, v) = A(\text{constant})$$

$$G(u, v) = H(u, v)F(u, v)$$

$$\Rightarrow H(u, v) = \frac{G(u, v)}{A}$$

$G(u, v)$ is the Fourier transform of the observed image and A is a constant describing the strength of the impulse.

9.4.3 Estimation by Mathematical modeling:

A degradation model proposed by Hufnagel and Stanley [1964] is based on the physical characteristics of atmospheric turbulence.

$$H(u, v) = e^{-k(u^2 + v^2)^{5/6}} \quad \begin{array}{l} \rightarrow \text{atmospheric turbulence} \\ k \rightarrow \text{Nature of the turbulence} \end{array}$$

Basic principles:

➤ What will be the degradation function where images are degraded by linear motion

Let, $f(x, y) \rightarrow$ motion

$x_0(t)$ and $y_0(t) \rightarrow$ time varying components

$T \rightarrow$ Shutter open (exposure) time

- Ex. Atmospheric model $H(u, v) = e^{-k(u^2 + v^2)^{5/6}}$

original



k=0.0025



k=0.001



k=0.00025



Total exposure at any point, (x, y)

$$g(x, y) = \int_0^T f(x - x_0(t), y - y_0(t)) dt$$



Observed blurred image

The Fourier transform is

$$\begin{aligned} G(u, v) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) e^{-j2\pi(ux+vy)} dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left[\int_0^T f(x - x_0(t), y - y_0(t)) dt \right] e^{-j2\pi(ux+vy)} dx dy \end{aligned}$$

$$G(u, v) = \int_0^T \underbrace{\left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x - x_0(t), y - y_0(t)) e^{-j2\pi(ux+vy)} dx dy \right]}_{F.T \text{ of shifted } f(x,y)} dt$$

Using this expression, we can write

$$f[x - x_0(t), y - y_0(t)] \Leftrightarrow F(u, v) e^{-j2\pi[ux_0(t) + vy_0(t)]} dt$$

Shift property of F.T

$$G(u, v) = \int_0^T F(u, v) e^{-j2\pi [ux_0(t) + vy_0(t)]} dt$$

$$= F(u, v) \int_0^T e^{-j2\pi [ux_0(t) + vy_0(t)]} dt$$

if you define degradation function $H(u, v)$

$x_0(t) \rightarrow$
 $y_0(t) \rightarrow$

} *Motion variables are known*

$$G(u, v) = H(u, v) F(u, v)$$